MILITARY SPECIFICATION

RECEIVER, RADIO R-1047A/A

This specification has been approved by the
Bureau of Naval Weapons, Department of the Navy

1. SCOPE

1.1 Scope - The equipment covered by this specification shall be of one type and shall provide for the reception of signals from radio sonobuoys to provide sonobuoy homing and on-top position indicating when used in conjunction with direction finder equipment.

2. APPLICABLE DOCUMENTS

2.1 Applicable Documents - The following documents of the issue in effect on the date of invitation for bids form a part of this specification to the extent specified herein:

SPECIFICATIONS

Military

MIL-P-116	Preservation, Methods of
MIL-E-4682	Electron Tubes and Transistors, Choice and Application of
MIL-W-5088	Wiring, Aircraft, Installation of
MIL-E-5400	Electronic Equipment, Aircraft General Specification for
MIL-T-5422	Environmental Testing, Aircraft Electronic Equipment
MIL-I-6181	Interference, Controlled Requirements, Aircraft Equipment
MIL-E-17555	Electronic and Electrical Equipment and Associated Repair Parts, Preparation for Delivery of

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MIL-T-18303	Test Procedures: Preproduction and Inspection for Aircraft Electronic Equipment, Format for
MIL-N-18307	Nomenclature and Nameplates for Aeronautical Electronic and Associated Equipment
MIL-R-23094	Reliability Assurance for Production Acceptance of Avionic Equipment, General Specification for

STANDARDS

Military

MIL-STD-704	Electric Power, Aircraft, Characteristics and Utilization of
MS-25245	Connections, Input Power Airborne Electronic Equipment; 28-Volt DC, 115-Volt Single Phase AC, or 115/200 Volt 3 Phase AC, or Combinations

2.2 Avialability of Documents -

(1) When requesting specifications, standards, drawings and publications, refer to both title and number. Copies of this specification and applicable specifications required by contractors in connection with specific procurement functions may be obtained upon application to the Commanding Officer, Naval Supply Depot, Code CDS, 5801 Tabor Avenue, Philadelphia 20, Pennsylvania.

3. REQUIREMENTS

3.1 Parts and Materials - In the selection of parts and materials, fulfillment of major design objectives shall be the prime consideration. In so doing the following shall govern:

(1) Parts and materials requirements shall conform to Specification MIL-E-5400.

(2) Non-repairable subassemblies, as outlined in Specification MIL-E-5400, shall be used when practicable. The general size of the subassembly and the amount of circuitry to be included therein shall be approved by the procuring activity. Non-repairable subassemblies must be reliable.

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(3) When previously produced models of this equipment did not use non-repairable subassemblies, the design shall not be changed to employ non-repairable assemblies without the approval of the procuring activity.

3.1.1 Non-Standard Parts and Material Approval - Approval for the use of non-standard parts and materials shall be obtained as outlined in Specification MIL-E-5400.

3.1.2 Electron Devices - Electron tubes, transistors and diodes shall be chosen and applied and the complements reported as outlined in Specification MIL-E-4682. The complement report must be submitted to the procuring activity for review and approval prior to preproduction testing.

3.2 Design and Construction - The equipment shall conform to all the applicable requirements of Specification MIL-E-5400 for design, construction and workmanship, unless otherwise specified herein.

3.2.1 Form Factor - The equipment shall be enclosed in a single case not to exceed 3.0 inches wide by 4.88 inches high by 5.25 inches long. The receiver shall be hard mounted without vibration isolators.

3.2.2 Total Weight - The total weight of the equipment, excluding cables and mount, shall be a minimum consistent with good design and shall not exceed 5 pounds.

3.2.3 Reliability -

3.2.3.1 Operational Stability - The equipment shall operate with satisfactory performance, continuously or intermittently, for a period of at least 400 hours without necessity for readjustment of any controls which are inaccessible to the operator during normal use.

3.2.3.2 Operating Life - The equipment shall have a minimum total operating life of 5000 hours with reasonable servicing and replacement of parts. Parts requiring scheduled replacement due to wear during the life of the equipment shall be determined by the contractor and submitted to the procuring agency.

3.2.3.3 Reliability in Mean Time Between Failures (MTBF) - For the test program under 4.4.3 the MTBF shall be 400 hours. Procedure II of Specification MIL-R-23094 shall be used.

3.2.4 Controls - All operational control of the receiver shall be from a separate radio set control.

3.2.5 Cabling and Connections -

3.2.5.1 Cables and Connectors - The equipment shall provide for the use of cables and connectors in accordance with Specification MIL-E-5400.

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3.2.5.2 Interconnection Cabling - The equipment shall be capable of satisfactory operation using external wiring in accordance with the applicable requirements of Specification MIL-W-5088. The external wiring shall be unshielded, except that a minimum number of the individual wires may be shielded when demonstrated as necessary to meet interference control requirements and provided the assembly of the cable to its plugs may be easily accomplished. External cables and that portion of the connectors attached to the cables shall not be supplied as part of the equipment. Antenna sharing capability shall be provided by an internal miniature coaxial relay. Relaying of tube plate power for the direction finder group shall be provided internally. Associated system functions including AGC, sensitivity, audio and ADF shall also be switched as necessary. All receiver connectors as well as all circuit protective devices, if required, shall be located on the front of the unit.

3.2.6 Interchangeability - The equipment shall meet the interchangeability requirements as defined in Specification MIL-E-5400.

3.2.7 Interference Control - The generation of radio interference by the equipment and the vulnerability of the equipment to radio interference shall be controlled within the limits of Specification MIL-I-6181.

3.2.8 Maintenance Provisions and Field Testing - Provisions for maintenance shall be as specified in Specification MIL-E-5400. Test points shall be provided on the front of the receiver to monitor significant voltages necessary to the maintenance cycle, including as a minimum:

a. Direction Finder Group Power Supply
b. Receiver Power Supply
c. Audio Output
d. AGC
e. Ground

Additional internal test points as necessary shall be provided for ease of trouble analysis.

3.2.9 Nomenclature and Nameplates - Nomenclature assignment and nameplate approval for equipment identification shall be in accordance with Specification MIL-N-18307.

3.2.10 Standard Conditions - The following conditions shall be used to establish normal performance characteristics under standard conditions and for making laboratory bench tests.

Temperature	Room ambient ($30^{\circ} \pm 10^{\circ}C$)
Altitude	Normal ground
Vibration	None

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Humidity — Room ambient up to 90% relative humidity

Input power voltage 115 ± 1.0 V AC, 400 cps and/or 27.5 ± 0.5 V DC

3.2.11 Service Conditions - The equipment shall operate satisfactorily under any of the environmental service conditions or reasonable combination of these conditions as specified in Specification MIL-E-5400 for Class 1 equipment, except as modified herein.

3.2.11.1 Vibration - The equipment shall operate satisfactorily when subjected to the vibration requirements of Curve I of Specification MIL-T-5422.

3.2.11.2 Altitude - The receiver shall be capable of satisfactory operation at altitudes up to 35,000 feet.

3.2.12 Warm-Up Time - The time required for the equipment to warm up prior to operations shall be kept to a minimum, and shall not exceed 5 seconds under standard conditions and 15 seconds at extreme service conditions.

3.2.13 Electric Power -

3.2.13.1 Input - The equipment shall conform to MIL-STD-704, and shall meet all other requirements of this specification when furnished with the following electric power as defined in MIL-STD-704:

Kind of Power: 115 volt, 400 cycle; 115 volt-amps; Category B
28 volts DC; .75 amperes continuous, 5 amperes during channel selection; Category B

3.2.13.2 Output - The receiver shall be capable of continuously providing 225 volts DC ± 10 per cent at 30 milliamperes ± 20 per cent tube plate output power for either the AM-608/ARA-25 or AM-1777/ARA-25A Electronic Control Amplifiers. This voltage shall contain no more than 0.50 volts ripple (RMS).

3.2.14 Internal Adjustments - Adjustments shall be kept to a minimum and shall be easily accessible upon removal of receiver dust cover. All adjustments shall be factory aligned and preset prior to delivery so that the receiver is operating satisfactorily to the requirements herein. Alignment shall be stable for a minimum of 400 normal operation hours.

3.2.15 Dust Cover - An external dust cover shall be supplied which is readily removable without attachment to wiring or other obstructions. Dust cover shall provide adequate sand and dust protection as specified by MIL-T-5422.

3.3 Performance - Unless otherwise specified, values set forth to establish the requirements for satisfactory performance apply to

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performance under both standard and extreme service conditions. When reduced performance under the extreme conditions is acceptable, tolerances or values setting forth acceptable variations from the performance under the standard conditions will be specified.

3.3.1 Operation - The receiver shall be compatible with, match impedance with and accept the voltage and impedance output levels from the AS-578/ARA-25 or AS-578A/ARA-25 antenna assemblies. The receiver shall preamplify, convert, I.F. amplify and amplitude detect the antenna output and provide acceptable voltage and impedance output levels to be compatible with the input requirements of the Electronic Control Amplifiers AM-1777/ARA-25A or AM-608/ARA-25. The receiver shall operate in conjunction with the AN/ARA-25 Direction Finder Group to provide a sonobuoy homing capability as displayed on an appropriate bearing, heading indicator. The AN/ARA-25 power supply shall be contained as a separate module, and shall be removable without affecting the overall receiver function.

3.3.2 Frequency Range - The receiver shall be designed to operate over the range of frequencies from 162 to 174 mc in such manner that any one of the following thirty-one channels may be switch-selected.

Channel No.	Frequency (MC)	Channel No.	Frequency (MC)
1	162.250	17	162.625
2	163.000	18	163.375
3	163.750	19	164.125
4	164.500	20	164.875
5	165.250	21	165.625
6	166.000	22	166.375
7	166.750	23	167.125
8	167.500	24	167.875
9	168.250	25	168.625
10	169.000	26	169.375
11	169.750	27	170.125
12	170.500	28	170.875
13	171.250	29	171.625
14	172.000	30	172.375
15	172.750	31	173.125
16	173.500		

3.3.3 Local Oscillator - The local oscillator stability shall be such that any change in alignment, including reset error caused by cycling of the channel selector or drift caused by aging of component parts, shall not exceed ± 10 kilocycles from the assigned channel frequency for the temperature range from -20°C to +35°C. For the temperature ranges from -54°C to -20°C and from +35°C to +55°C, the local oscillator drift shall not exceed ± 15 kilocycles from the assigned channel frequency.

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3.3.4 Overall Gain Stability - The gain stability of this receiver shall be such that regenerative oscillatory conditions do not prevail for any value of carrier level at the antenna connector below 10,000 microvolts.

3.3.5 Sensitivity - A 10-decibel (db) signal-to-noise ratio shall exist at the audio output when a 2.5-microvolt signal, 30 per cent modulated by a 100 cps square wave is applied at the antenna connector.

3.3.6 Noise Figure - The noise figure of this equipment shall be no greater than 12 db. A nominal value shall be 6 db.

3.3.7 Single-Channel Bandwidth - For any given channel, the receiver minimum 6 db bandwidth shall be no less than 190 kilocycles (kc) and no more than 255 kc. The slope factor shall provide a minimum of 60 db adjacent channel rejection.

3.3.8 IF Rejection - The receiver shall attenuate the IF center frequency at least 80 db relative to any desired channel center frequency applied at the receiver input.

3.3.9 Image and Other Spurious Responses - The receiver shall attenuate image and other spurious responses external to the passband at least 60 db compared to any desired channel center frequency except at $\frac{1}{2}$ signal frequency the response shall be down at least 45 db. There shall be no more than two spurious responses in any channel within the passband which are less than 60 db down, and these shall be at least 50 db down.

3.3.10 RF Input Protection - The receiver shall be suitably protected for input signals up to 500,000 microvolts at any frequency or combination of frequencies below 30,000 mc.

3.3.11 Audio Output Characteristics -

3.3.11.1 Frequency Response - The frequency response of the receiver shall be such that the audio output from 75-500 cps is flat within $\pm$ 2 db of the reference level output at 100 cps when measured over an external 22,000 ohm resistive impedance.

3.3.11.2 Audio Level Nominal - The nominal root-mean-square (rms) level of audio output for 100 cps shall be 0.5 volts.

3.3.11.3 Audio Level Range - When a 100 cps, 30 per cent square wave modulated carrier signal of 2.5 microvolts or greater is applied to the receiver, the amplitude detected and amplified audio output shall be adjustable from .25 volts rms minimum to 1.0 volt rms over an external 22,000 ohm impedance.

3.3.12 Overall Phase Delay - Phase delay between the 100 cps modulation entering the receiver and the 100 cps demodulated audio signal

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used as the input to the Electronic Control Amplifiers of the AN/ARA-25 Direction Finder Group shall be 180 ± 10 degrees.

3.3.13 RF Bandpass Filter - RF Amplifier Circuits shall feature an input and output tuned circuit or filter to reject signals below 162.25 megacycles and above 173.50 megacycles.

3.3.14 Reduced Size Coaxial Relay - Internal coaxial relay used for antenna sharing shall have a maximum of 0.5 db insertion loss and shall have a minimum of 40 db isolation between open contacts measured at 225 megacycles.

3.3.15 Remote "Adequate Signal Strength Indicator" - The receiver shall contain a two-pole relay and a threshold control that is adjustable and shall operate upon receiving an RF signal of -86 dbm at the receiver antenna connector. It shall never operate in the absence of signal. When actuated, the relay shall supply a ground to the external circuit. Depending upon circuit design, performance of the above function without the use of a threshold control is preferred.

3.3.16 Headphone Output - A nominal audio output of 4 to 6 volts (rms) shall be supplied at 400 cycles. This audio shall be keyed by one pole of the two-pole relay used in 3.3.15. The output impedance of the audio line shall be 600 ohms.

3.3.17 Radio Interference Suppression - Radio interference suppression shall conform to the requirements of MIL-I-6181. The receiver shall include radio frequency (RF) bandpass features in the RF amplifier and adequate individual channel selectivity in succeeding stages.

4. QUALITY ASSURANCE PROVISIONS

4.1 General - Items covered by this specification shall be subjected to the following tests to determine compliance with all applicable requirements:

(1) Preproduction Tests

(2) Production Sample Tests

(3) Acceptance Tests

(4) Life Tests

4.1.1 Inspection Responsibility - The contractor is responsible for the performance of all inspection requirements specified. The contractor may use his or other inspection facilities and services that are acceptable to the Government. Inspection records of tests and product examination shall

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be kept complete, and shall be available to the Government as specified in the contract or order. The Government reserves the right to perform any inspections deemed necessary to ensure that supplies and services conform to prescribed requirements.

4.1.2 Test Conditions -

4.1.2.1 Electric Power - For all tests requiring operation of the equipment, the equipment shall be provided with electric power in accordance with 3.2.13.1.

4.2 Preproduction Tests - Preproduction tests shall be made on an equipment representative of the production equipments to be supplied under the contract. Preproduction tests shall consist of the following:

(1) Contractor's Demonstration Tests

(2) Service Approval Tests

4.2.1 Contractor's Demonstration Tests - Contractor's demonstration tests shall be accomplished under the responsibility of the contractor and shall be conducted in accordance with the approved test procedure of 4.6. The government inspector and the procuring activity shall be advised when tests are to be conducted so that a representative may be designated to witness or supervise the tests when so desired. Contractors not having adequate facilities to conduct all required tests shall obtain the services of a commercial testing laboratory.

4.2.1.1 Extremes of Potential and Frequency - The equipment shall be subjected to operating tests with the electric power having combinations of applicable frequency and voltage extremes permitted by MIL-STD-704 for the specified conditions.

4.2.1.2 The contractor shall submit all data collected in conducting these tests to the procuring agency for review and approval. This data shall include a list of all electrical and electronic parts giving their specified voltage, current and temperature rating and the applied circuit voltage, current and ambient and surface temperatures. The ambient and surface temperature shall be obtained under the extreme high temperature operating condition. Such data on parts may be included as part of the preliminary parts list.

4.2.2 Service Approval Tests - At the completion of the contractor's demonstration tests and when requested by the procuring activity, the equipment shall be delivered to a specified government laboratory for additional testing. This additional testing may consist of duplicating tests previously conducted and such other tests as are deemed necessary to determine compliance with all applicable design and

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performance requirements.

4.2.2.1 Accessory Material - In addition to the complete equipment submitted for service approval tests, the contractor shall also submit the accessory material and design and test data specified in Specification MIL-E-5400. This information shall indicate the physical and electrical characteristics of the equipment and establish the equipment's compliance with applicable requirements.

4.2.3 Scope of Tests - Preproduction tests shall include all tests deemed necessary to determine that the equipment meets all the requirements of this specification and the contract. Preproduction tests shall include environmental tests in accordance with Specification MIL-T-5422 and interference tests in accordance with Specification MIL-I-6181.

4.2.4 Preproduction Approval - Approval of the preproduction model shall be by the procuring activity upon satisfactory completion of all tests. No production equipments shall be delivered prior to the approval of the preproduction model. Prefabrication of production equipment prior to the approval of the preproduction model is at the contractor's own risk. The approved preproduction model will be returned to the contractor for his use in the fabrication and testing of equipment to be submitted for acceptance. The preproduction model shall not be considered as one of the equipments under the contract; however, it may be reworked by the contractor and submitted for acceptance as a production equipment.

4.2.5 Equipments supplied under the contract shall in all respects, including design, construction, workmanship, performance and quality, be equivalent to the approved preproduction sample. Each equipment shall be capable of successfully passing the same tests imposed on the preproduction sample. Evidence of non-compliance with the above shall constitute cause for rejection and for equipment already accepted by the government it shall be the obligation of the contractor to make necessary corrections as approved by the procuring activity.

4.3 Production Sample Tests - When requested by the procuring activity, one equipment shall be selected from the first 10 production equipments submitted for acceptance and sent to a designated government laboratory for tests. The equipment shall be selected by the government inspector after the equipment has successfully passed all individual tests.

4.3.1 Scope of Tests - This equipment may be subjected to any and all tests the procuring activity deems necessary to ensure that the production equipment is equivalent to the previously approved preproduction sample in design, construction, workmanship, performance and quality and that it meets all applicable requirements.

4.3.2 Production Sample Approval - Approval of the production sample shall be by the procuring activity upon satisfactory completion of

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all tests. Any design, material or performance defect made evident during this test shall be corrected by the contractor to the satisfaction of the procuring activity. Failure of the production sample to pass any of the tests shall be cause for deliveries of equipment under the contract to cease until proper corrective action is approved and accomplished. Corrective action shall also be accomplished on equipment previously accepted when requested by the procuring activity.

4.3.3 Reconditioning of Production Test Sample - Upon completion of the production sample test the equipment shall be reworked by the contractor by replacing all worn or damaged items. After reworking the contractor shall resubmit the equipment for acceptance.

4.4 Acceptance Tests - The contractor shall furnish all samples and shall be responsible for accomplishing the acceptance tests. All inspection and testing shall be under the supervision of the government inspector. Contractors not having testing facilities satisfactory to the procuring activity shall engage the service of a commercial testing laboratory acceptable to the procuring activity. The contractor shall furnish test reports showing quantitative results for all acceptance tests. Such reports shall be signed by an authorized representative of the contractor or laboratory, as applicable. Acceptance or approval of material during the course of manufacture shall not be construed as a guarantee of the acceptance of the finished product. Acceptance tests shall consist of the following:

(1) Individual Tests

(2) Sampling Tests

(3) Reliability Assurance Tests

(4) Special Tests

4.4.1 Individual Tests - Each equipment submitted for acceptance shall be subjected to the individual tests. These tests shall be adequate to determine compliance with the requirements of material, workmanship, operational adequacy and reliability. As a minimum, each equipment accepted shall have passed the following tests:

(1) Examination of Product

(2) Operational Test

(3) Manufacturing Run-in Test

4.4.1.1 Examination of Product - Each equipment shall be examined carefully to determine that the material and workmanship requirements have been met.

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4.4.1.2 Operational Test - Each equipment shall be operated long enough to permit the equipment temperature to stabilize and to check sufficient characteristics and record adequate data to ensure satisfactory equipment operation.

4.4.1.3 Manufacturing Run-in Test - Each equipment shall be operated under the conditions specified herein for a period of 6 hours without failure. A failure shall be defined as anything which causes malfunctioning of the equipment. Only those adjustments will be permitted which can be made by using such controls and adjustments that are accessible to the operator during the normal use of the equipment.

Temperature	Ambient room
Humidity	Ambient room
Vibration	Any selected frequency within the range of 20 to 30 cps (excluding resonant points) and a minimum amplitude of ±3g's

The equipment shall be vibrated (without vibration isolators) for a period of 10 minutes prior to the beginning of the 6-hour period of operation. Where feasible, the equipment shall be operated during this vibration period for the purpose of detecting flaws and imperfect workmanship. Operation within the specified limits of satisfactory performance is not necessarily required during the vibration period. The direction of vibration should be vertical to the normal mounting plane for 5 minutes and lateral to that plane for 5 minutes. Where it is not feasible to vibrate the equipment in 2 directions the vertical direction shall be used. During the 6-hour period of operation following the 10-minute vibration period, the equipment shall be mechanically cycled periodically through its various phases of operation. Should a failure occur, it should be repaired and the test started over, except that the 10-minute vibration period need not be repeated when it is certain the failure was not a result of the vibration. Should repetitive failures occur, corrective action shall be taken to eliminate this defect from future equipment. A record shall be kept of all failures. The 6-hour period specified above may be composed of two 3-hour periods to conform with standard working hours.

4.4.2 Sampling Tests - Sampling tests shall be conducted on equipments which have successfully passed the individual tests. Equipments shall be selected for sampling tests by the Government Inspector in accordance with the following:

Quantity of Equipments Offered for Acceptance	Quantity to be Selected For Sampling Test
First 10	1
Next 50	1

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Next 75	1
Next 100	1
	1 for each additional 200 or fraction thereof

When a defective unit is found, acceptance of any other units in the group shall be withheld until the extent and cause of failure have been determined and appropriately corrected. Return to one sample out of the first ten items and proceed as indicated when inspection is resumed.

4.4.2.1 Scope of Tests - As a minimum, each equipment selected for sampling tests shall be subjected to the following tests:

(1) Complete operational test at ambient room conditions, making all necessary measurements to ensure that all applicable specification requirements have been met.

(2) Operational test at certain environmental conditions. The conditions may vary for each equipment tested and should be based on results of the preproduction, production and special tests.

(3) Manufacturing run-in test specified in 4.4.1.3 except that the test duration shall be 120 hours with no restriction on the number of failures. However, each failure shall be analyzed as to cause and remedial action necessary to reduce the possibility of its recurrence in future equipment.

4.4.3 Reliability Assurance Tests - Reliability assurance tests shall be conducted as required by specification MIL-R-23094. Equipments selected for reliability assurance tests shall first have passed the individual tests.

4.4.3.1 Reliability Procedure - Reliability Procedure II from Specification MIL-R-23094 shall be used.

4.4.3.2 Test Level - The test level shall be level III from Table I of Specification MIL-R-23094.

4.4.3.3 Accept-Reject Criteria - Figure 11 of Specification MIL-R-23094 shall be used to determine the accept-reject criteria for qualification and Figure 12 shall be used for reliability sampling tests.

4.4.3.4 Performance Characteristics to be Measured - The equipment shall meet the performance characteristics described in 3.3.

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4.4.3.5 Failure Criteria - In addition to the requirements of Specification MIL-R-23094, the following requirements shall be used to determine when a failure has occurred during the test:

(1) Whenever performance characteristics fall below the acceptance requirement at least one failure has occurred. If subsequent analysis reveals that several parts deteriorated, each shall be counted a failure, unless one caused the other parts to fail.

4.4.3.6 Preventive Maintenance - During the period of the tests such maintenance procedures may be performed as are possible for the operator during normal use of the equipment and as are described in the maintenance instructions. Such procedures shall be approved by the procuring activity prior to acceptance for reliability assurance tests.

4.4.4 Special Tests - Special tests shall be conducted on a quantity of equipments for the purpose of checking the effect of any design or material change on the performance of the equipment and to assure adequate quality control. The equipment selected for special tests may be selected from equipments previously subjected to the sampling tests.

4.4.4.1 Special Tests Schedule - Selection of equipments for special tests shall be made as follows, except that when a Production Sample is requested and delivered to a government laboratory for test (see 4.3) the requirement for a special test on an early production equipment may be omitted:

(1) On an early production equipment.

(2) On an early equipment after an engineering change.

(3) Whenever failure reports or other information indicate additional equipments should be tested. (This will be determined by the procuring activity.)

4.4.4.2 Scope of Tests - Special tests shall consist of such tests as are approved by the procuring activity. Test procedures previously approved for the preproduction and sampling tests shall be used where applicable. When not applicable, the contractor shall prepare a test procedure and submit it to the procuring activity for approval prior to conducting the tests. As a minimum, the following tests shall be made, except that for those equipments selected as a result of design changes and failure reports, only those tests required to check the characteristics in question need be conducted:

(1) Temperature - Altitude Tests

(2) Vibration Test (60 minutes cycling in each of 3 planes)

(3) Humidity Tests

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4.4.5 Equipment Failure - Should a failure occur during either the sampling or special tests, the following action shall be taken:

(1) Determine the cause of failure.

(2) Determine if the failure is an isolated case or design defect.

(3) Submit to the procuring activity for approval, proposed corrective action intended to reduce the possibility of the same failure(s) occurring in future tests.

(4) Where practicable, include a test in the individual test to check all equipment for this requirement until reasonable assurance is obtained that the defect has been satisfactorily corrected.

4.5 Life Test - The contractor shall furnish all samples and shall be responsible for accomplishing the life test. The test shall be of 300 hours duration and shall be conducted on equipments that have passed the individual test. The life test shall be performed under the conditions specified in 4.5.1. The life test sample shall be selected by the government inspector in accordance with the following, except that when a production sample equipment is requested and submitted to the procuring activity for test the first equipment specified below need not be selected or subjected to this test: (Equipments which have successfully passed the sampling tests or special tests may be selected for life tests.)

Quantity of Equipments Offered for Acceptance	Quantity to be Selected For Life Test
First 25	1
Next 175	1
Next 300	1
	1 for each additional 500 or fraction thereof

When a defective unit is found, acceptance of any of the units in the group shall be withheld until the extent and cause of failure have been determined and appropriately corrected. Return to one sample out of the first 25 items and proceed as indicated when inspection is resumed.

4.5.1 Test Conditions - The life test shall be conducted under the following simulated service conditions:

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Temperature	Normal room
Altitude	Normal ground (0-5000 ft.)
Humidity	Room ambient
A. C. Voltage	115 ± 5 volts (at lowest applicable frequency)
D. C. Voltage	27.0 ± 2.0 volts

4.5.2 Test Periods - The test may be run continuously or intermittently. Any period of operation shall be of sufficient duration to permit the equipment temperature to stabilize. Periodically, the equipment shall be turned on and off several times and put through its various phases of operation.

4.5.3 Performance Check - At approximately 8-hour intervals during the test, a limited performance check shall be made. The performance check proposed by the contractor shall be subject to approval by the procuring activity.

4.5.4 Test Data - The contractor shall keep a daily record of the performance of the equipment, making particular note of any deficiencies or failures. In the event of part failures, the defective part shall be replaced and the operation resumed for the balance of the test period. A record shall be kept of all failures throughout the test, including all tube failures. This record shall indicate the following:

(1) Part type number

(2) The circuit reference symbol number

(3) The part function

(4) Name of the manufacturer

(5) Nature of the failure

(6) The number of hours which the part operated prior to failure.

4.5.4.1 Failure Report - In the event of a failure, the government inspector shall be notified immediately. A report shall be submitted to the procuring activity upon completion of the test. In this report, the contractor shall propose suitable and adequate design or material corrections for all failures which occurred. The procuring activity will review such proposals and determine whether they are acceptable.

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4.5.5 Reconditioning of Life Test Samples - An equipment which has been subjected to the life test shall be reconditioned as follows:

(1) On completion of the life test, the equipment shall be reworked by the contractor by replacing all "wear" items. The "wear" items shall be determined by agreement between the contractor and the procuring activity.

(2) After reworking, the contractor may resubmit the equipment for acceptance.

4.6 Test Procedures - The procedures used for conducting preproduction tests, acceptance tests and life tests shall be prepared by the contractor and submitted to the procuring activity for review and approval. The right is reserved by the procuring activity or the government inspector to modify the tests or require any additional tests deemed necessary to determine compliance with the requirements of this specification or the contract. Specification MIL-T-18303 shall be used as a guide for preparation of test procedures. When test procedures are available from previous contracts such procedures may be used when approved by the procuring activity.

4.7 Presubmission Testing - No item, part or complete equipment shall be submitted by the contractor until it has been previously tested and inspected by the contractor and found to comply, to the best of his knowledge and belief, with all applicable requirements.

4.8 Rejection and Retest - Equipment which has been rejected may be reworked or have parts replaced to correct the defects and may be resubmitted for acceptance. Before resubmitting, full particulars concerning previous rejection and the action taken to correct the defects found in the original shall be furnished the government inspector.

5. PREPARATION FOR DELIVERY

5.1 General - All major units and parts of the equipment shall be preserved, packaged, packed and marked for the level of shipment specified in the contract or order in accordance with Specification MIL-E-17555. For Level A packaging, the method of packaging shall conform to Method II of MIL-P-116.

6. NOTES

6.1 Intended Use - This equipment will be used in Navy ASW aircraft to provide sonobuoy homing and on-top position indicating. It will serve the same purpose as Receiver, Radio R-1047/A with 31 channels of radio frequency vice the 16 channel capability of the Receiver, Radio R-1047/A. The additional 15 channel center frequencies are interspaced at increments of 375 kc midway between the existing 16 channel center frequencies.

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6.2 Test Values - Normal and limiting values of performance data shall be determined at input voltages of 27.5 ± 0.5 V DC and 115 ± 1.0 V AC as applicable. These data are to be used in testing the equipment at installation points for compliance with minimum acceptable standard of performance.

6.3 Performance Objectives - Minimum size and weight, simplicity of operation, ease of maintenance, and an improvement in the performance and reliability of the specific functions beyond the requirements of this specification are objectives which shall be considered in the production of this equipment. Where it appears a substantial reduction in size and weight or improvement in simplicity of design, performance, ease of maintenance or reliability will result from the use of materials, parts and processes other than those specified in Specification MIL-E-5400, it is desired their use be investigated. When investigation shows advantages can be realized, a request for approval shall be submitted to the procuring activity for consideration. Each request shall be accompanied by complete supporting information.

6.4 Non-Repairable Subassemblies - As a general rule, non-repairable subassemblies should be encapsulated or hermetically sealed. The number of connections internal to the subassembly should be held to a minimum. Detail parts tolerances and ratings should be so selected that the life of the subassembly is greater than that of a similar repairable one. With few exceptions (such as high voltage power supplies), the non-repairable subassembly should evidence a mean-time-to-failure greater than 5000 hours, and for many applications this figure must be nearer 50,000 hours.

6.5 Precedence of Documents - When the requirements of the contract, this specification, or applicable subsidiary specifications are in conflict, the following precedence shall apply:

(1) Contract - The contract shall have precedence over any specification.

(2) This Specification - This specification shall have precedence over all applicable subsidiary specifications. Any deviation from this specification, or from subsidiary specifications where applicable, shall be specifically approved in writing by the procuring activity.

(3) Referenced Specifications - Any referenced specification shall have precedence over all applicable subsidiary specifications referenced therein. All referenced specifications shall apply to the extent specified.

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NOTICE - When Government drawings, specifications, or other data are used for any purpose other than in connection with a definitely related Government procurement operation, the United States Government thereby incurs no responsibility nor any obligation whatsoever; and the fact that the Government may have formulated, furnished or in any way supplied the said drawings, specifications or other data is not to be regarded by implication or otherwise as in any manner licensing the holder or any other person or corporation, or conveying any rights or permission to manufacture, use or sell any patented invention that may in any way be related thereto.

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SPECIFICATION ANALYSIS SHEET
NAVWEPS FORM 4121/3 (8-62)

FORM APPROVED
BUDGET BUREAU NO. 45-R309

INSTRUCTIONS

This sheet is to be filled out by Government or contractor personnel involved in the use of this specification in procurement of products for ultimate use by the activity shown on the reverse of this sheet.

This sheet is provided for obtaining information which will aid the activity shown in insuring that through use of this specification suitable products can be procured with a minimum amount of delay and at the least cost.

Comments and the return of this sheet will be appreciated.

Fold on dotted lines on reverse side, staple in corner, and send to the activity shown.

SPECIFICATION *(No. and abbrev. title)*

ORGANIZATION	CITY	STATE

CONTRACT NUMBER	QUANTITY OF ITEMS PROCURED	DOLLAR AMOUNT
		$

MATERIAL PROCURED UNDER A DIRECT GOVERNMENT CONTRACT ☐ OR A SUBCONTRACT ☐

1. HAS ANY PART OF THIS SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?
(If so, give paragraph number and wording, and recommendations for correcting the deficiencies.)

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?
☐ YES ☐ NO IF YES, IN WHAT WAY?

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If additional papers are attached, send this form and papers in an envelope. This form is addressed for use in window envelope when appropriate.)*

SUBMITTED BY *(Print name and title)*	DATE

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